Title: The Edge-cloud Continuum Meets Log-structured Storage Engines: A Framework for Energy-efficient Resource Allocation

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Authors: T. Banerjee, I. Fernandez, E. Haddad (Centre for Secure Systems, Eastvale Institute of Technology)
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Abstract:

Energy-efficient resource allocation remains a critical challenge for the edge-cloud continuum, particularly in enterprise resource planning. This paper proposes a framework built on log-structured storage engines that explicitly accounts for legacy system migration. We evaluate the framework on the Google cluster trace and compare it against 4 baseline methods. Experimental results show that the proposed approach improves 99th-percentile latency by 10.4% while keeping defect detection rate within acceptable bounds. We further analyse its limitations and discuss implications for practical deployment.

1. Introduction

The rapid maturation of the edge-cloud continuum has changed how researchers and practitioners approach energy-efficient resource allocation, yet practical deployment is still limited by scalability, cost, and reproducibility concerns. Over the past decade, log-structured storage engines have attracted sustained attention as a promising route toward solving energy-efficient resource allocation in realistic settings such as enterprise resource planning. Despite considerable progress, existing solutions for energy-efficient resource allocation tend to degrade when moving from controlled laboratory conditions to production workloads, which motivates a more systematic investigation. A central difficulty is that improvements in request throughput frequently come at the expense of defect detection rate, forcing designers to accept trade-offs that are poorly understood.

This paper addresses these gaps by studying how log-structured storage engines interact with software-defined networking controllers when applied to energy-efficient resource allocation. Our work is motivated by the growing demand for enterprise resource planning, where reliability and transparency requirements are especially strict. Understanding legacy system migration is equally important, because it shapes both the theoretical limits and the operational cost of any proposed approach. The remainder of this paper reviews related work, describes the proposed methodology, and concludes with a discussion of limitations and directions for future research.

2. Related Work

OConnor et al. (2016) introduced an early framework based on log-structured storage engines, reporting a 37.0% improvement in 99th-percentile latency over previous baselines. Subsequent work by Nakamura et al. (2012) extended this line of research to enterprise resource planning, although the evaluation was restricted to small-scale settings. A survey by Fernandez et al. (2010) categorised existing approaches to energy-efficient resource allocation into three families and highlighted the lack of standard benchmarks. Researchers working on the edge-cloud continuum have explored software-defined networking controllers as an alternative, with Haddad et al. (2013) noting that the approach scales poorly beyond 1200 components. Chen et al. (2012) analysed legacy system migration and showed that purely heuristic solutions cannot guarantee consistent resource utilization.

Several studies have criticised the reliance on synthetic workloads, arguing that results on the Google cluster trace may not generalise to enterprise resource planning. In contrast to these efforts, our work treats energy-efficient resource allocation and legacy system migration jointly rather than as independent concerns. Industry reports corroborate these academic findings, pointing to energy-efficient resource allocation as a

persistent obstacle for enterprise resource planning. Despite this body of literature, no prior study has systematically compared log-structured storage engines and software-defined networking controllers under identical experimental conditions.

3. Methodology

We formalise energy-efficient resource allocation as a constrained optimisation task in which the objective is to maximise 99th-percentile latency while keeping request throughput within a prescribed budget. The proposed architecture consists of four stages: data acquisition, preprocessing, core processing based on log-structured storage engines, and a validation layer that incorporates software-defined networking controllers. All experiments were conducted on the Google cluster trace, which provides 32 labelled instances or configurations covering a broad range of operating conditions. To ensure reproducibility, every experiment was repeated 5 times with independent random seeds, and we report mean values together with 95% confidence intervals.

Hyperparameters were selected through grid search over a validation split, and the final configuration was fixed before any evaluation on held-out data. The core routine applies log-structured storage engines to reduce the effect of legacy system migration, followed by a calibration step that stabilises 99th-percentile latency. We compare against 4 baseline methods, including a conventional approach built on software-defined networking controllers and two recent variants reported in the literature. The proposed design improves 99th-percentile latency by 10.4% relative to the strongest baseline and reduces defect detection rate by 25.4%, a statistically significant difference ($p < 0.05$).

Ablation experiments show that removing the software-defined networking controllers component reduces 99th-percentile latency by approximately 1.7%, confirming its contribution. A sensitivity analysis over 32 parameter settings shows that performance remains stable, with variation in request throughput below 8.0%. Threats to validity include the limited diversity of the Google cluster trace and the possibility that deployment in enterprise resource planning introduces conditions not captured in our setup.

4. Conclusion

This paper investigated energy-efficient resource allocation through the lens of log-structured storage engines, and demonstrated that a carefully designed pipeline can improve 99th-percentile latency by 10.4% on the Google cluster trace. The findings suggest that the edge-cloud continuum will continue to benefit from closer integration of log-structured storage engines and software-defined networking controllers, particularly in enterprise resource planning. Several limitations remain, including the restricted scale of our evaluation and the simplified treatment of legacy system migration. Future work will extend the framework to larger deployments, explore tighter theoretical guarantees, and release an open benchmark to facilitate fair comparison. We hope the evidence presented here encourages the community to adopt more rigorous and transparent evaluation practices for energy-efficient resource allocation.